

Nama: Garcia Fernanda Valenca Archadea

NIM: 254107020175

Kelas: 2A

Prodi: DIV – TEKNIK INFORMATIKA

JOBSHEET PRAKTIKUM

SETUP AWAL

1. Install PostgreSQL

- Download dan install PostgreSQL dari <https://www.postgresql.org/download/>
- Install pgAdmin (GUI tool) atau gunakan psql (command line tool)

2. Login ke Database

```
psql -U username -d database_name
```

3. Buat Database Baru:

```
-- 2. Buat database baru  
CREATE DATABASE company_db;  
\c company_db;
```

4. Buat Tabel-tabel:

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```
-- 3. Buat tabel-tabel
CREATE TABLE departments (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    location VARCHAR(100)
);

CREATE TABLE employees (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    salary NUMERIC(10,2),
    department_id INTEGER REFERENCES departments(id),
    hire_date DATE
);

CREATE TABLE projects (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    budget NUMERIC(15,2),
    department_id INTEGER REFERENCES departments(id)
);

CREATE TABLE employee_projects (
    employee_id INTEGER REFERENCES employees(id),
    project_id INTEGER REFERENCES projects(id),
    hours_worked NUMERIC(5,2),
    assignment_date DATE DEFAULT CURRENT_DATE,
    PRIMARY KEY (employee_id, project_id)
);
```

5. Insert Data Sample:

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```
-- Data departments
INSERT INTO departments (name, location) VALUES
('IT', 'Jakarta'),
('HR', 'Bandung'),
('Finance', 'Surabaya'),
('Marketing', 'Medan'),
('Operations', 'Yogyakarta');

-- Data employees
INSERT INTO employees (name, salary, department_id, hire_date) VALUES
('Budi Santoso', 8000000, 1, '2022-01-15'),
('Siti Rahayu', 7500000, 1, '2022-03-20'),
('Ahmad Fauzi', 9000000, 2, '2021-11-10'),
('Dewi Anggraini', 8500000, 2, '2022-02-28'),
('Rudi Hermawan', 9500000, 3, '2021-08-05'),
('Maya Sari', 8800000, 3, '2022-04-15'),
('Hendra Pratama', 8200000, 4, '2022-05-20'),
('Lina Marlina', 7800000, 4, '2022-06-10'),
('Joko Widodo', 9200000, 1, '2021-12-01'),
('Ratna Dewi', 8700000, 3, '2022-03-15'),
('Fajar Nugroho', 0, NULL, '2023-01-01'), -- Employee tanpa department
('Bambang Sutoyo', 6500000, 5, '2022-07-01');
```

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```
-- Data projects
INSERT INTO projects (name, budget, department_id) VALUES
('Website Development', 50000000, 1),
('Mobile App', 75000000, 1),
('Recruitment System', 30000000, 2),
('Payroll System', 45000000, 3),
('Marketing Campaign', 60000000, 4),
('Database Migration', 35000000, 1),
('Inventory System', 40000000, NULL), -- Project tanpa department
('CRM Implementation', 55000000, 5);

-- Data employee_projects
INSERT INTO employee_projects (employee_id, project_id, hours_worked, assignment_date) VALUES
(1, 1, 120.5, '2023-01-15'),
(1, 6, 80.0, '2023-02-01'),
(2, 1, 95.5, '2023-01-20'),
(2, 2, 150.0, '2023-02-15'),
(3, 3, 200.0, '2023-03-01'),
(4, 3, 180.5, '2023-03-05'),
(5, 4, 220.0, '2023-04-01'),
(6, 4, 190.5, '2023-04-05'),
(7, 5, 175.0, '2023-05-01'),
(8, 5, 160.5, '2023-05-10'),
(9, 2, 140.0, '2023-06-01'),
(9, 6, 90.5, '2023-06-15'),
(10, 4, 210.0, '2023-07-01'),
(11, 7, 100.0, '2023-07-15'),
(12, 8, 180.0, '2023-08-01');
```

Studi Kasus: Sistem Manajemen Perusahaan XYZ

Database perusahaan dengan berbagai entitas: departments, employees, projects, dan hubungan many-to-many antara employees dan projects.

Praktikum 1: SELECT dengan LIMIT/OFFSET

Konsep Dasar

- **LIMIT:** Membatasi jumlah baris yang ditampilkan
- **OFFSET:** Melewati sejumlah baris tertentu
- **Perbedaan dengan MySQL:** PostgreSQL menggunakan LIMIT x OFFSET y vs MySQL LIMIT y, x

Latihan Praktis

1. Basic SELECT dengan LIMIT

```
SELECT * FROM employees LIMIT 5;
```

2. SELECT dengan ORDER BY dan LIMIT/OFFSET

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```
SELECT name, salary FROM employees
ORDER BY salary DESC
LIMIT 3 OFFSET 2;
```

3. SELECT dengan WHERE dan LIMIT

```
SELECT * FROM employees
WHERE salary > 8000000
LIMIT 4;
```

Soal Latihan Praktikum 1

1. Tampilkan 5 employee dengan gaji terendah
2. Tampilkan halaman 2 data projects (3 data per halaman)
3. Tampilkan 3 employee yang dihire paling akhir

Query Mahasiswa:

```
SELECT * FROM employees
ORDER BY salary ASC
LIMIT 5;

SELECT * FROM projects
ORDER BY id
LIMIT 3 OFFSET 3;

SELECT * FROM employees
ORDER BY hire_date DESC
LIMIT 3;
```

Hasil Eksekusi:

1.

	123 id	A-Z name	123 salary	123 department_id	hire_date
1	11	Fajar Nugroho	0	[NULL]	2023-01-01
2	12	Bambang Sutoyo	6,500,000	5	2022-07-01
3	2	Siti Rahayu	7,500,000	1	2022-03-20
4	8	Lina Marlina	7,800,000	4	2022-06-10
5	1	Budi Santoso	8,000,000	1	2022-01-15

2.

	123 id	A-Z name	123 budget	123 department_id
1	4	Payroll System	45,000,000	3
2	5	Marketing Campaign	60,000,000	4
3	6	Database Migration	35,000,000	1

3.

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	123 id	AZ name	123 salary	123 department_id	hire_date
1	11	Fajar Nugroho	0	[NULL]	2023-01-01
2	12	Bambang Sutoyo	6,500,000	5	2022-07-01
3	8	Lina Marlina	7,800,000	4	2022-06-10

Praktikum 2: JOIN Operations

Jenis-Jenis JOIN

- **INNER JOIN:** Data yang match di kedua tabel
- **LEFT JOIN:** Semua data dari tabel kiri + match dari kanan
- **RIGHT JOIN:** Semua data dari tabel kanan + match dari kiri
- **FULL OUTER JOIN:** Semua data dari kedua tabel

Latihan Praktis

1. INNER JOIN

```
SELECT e.name, d.name as department, e.salary
FROM employees e
INNER JOIN departments d ON e.department_id = d.id;
```

2. LEFT JOIN

```
SELECT e.name, d.name as department
FROM employees e
LEFT JOIN departments d ON e.department_id = d.id;
```

3. RIGHT JOIN

```
SELECT d.name as department, COUNT(e.id) as employee_count
FROM departments d
RIGHT JOIN employees e ON d.id = e.department_id
GROUP BY d.name;
```

4. FULL OUTER JOIN

```
SELECT e.name as employee, d.name as department
FROM employees e
FULL OUTER JOIN departments d ON e.department_id = d.id;
```

5. Multiple JOIN

```
SELECT e.name, p.name as project, ep.hours_worked
FROM employees e
INNER JOIN employee_projects ep ON e.id = ep.employee_id
INNER JOIN projects p ON ep.project_id = p.id;
```

Soal Latihan Praktikum 2

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1. Tampilkan semua projects beserta department penanggung jawabnya
2. Tampilkan employee yang tidak memiliki department
3. Tampilkan department yang tidak memiliki employee
4. Tampilkan employee yang bekerja di project 'Website Development'

Query Mahasiswa:

```
SELECT p.name as project, d.name as department
FROM projects p
LEFT JOIN departments d ON p.department_id = d.id;

SELECT e.name, e.salary
FROM employees e
LEFT JOIN departments d ON e.department_id = d.id
WHERE d.id IS NULL;

SELECT d.name as department
FROM departments d
LEFT JOIN employees e ON d.id = e.department_id
WHERE e.id IS NULL;

SELECT e.name, p.name as project, ep.hours_worked
FROM employees e
INNER JOIN employee_projects ep ON e.id = ep.employee_id
INNER JOIN projects p ON ep.project_id = p.id
WHERE p.name = 'Website Development';
```

Hasil Eksekusi:

1.

	AZ project	AZ department
1	Website Development	IT
2	Mobile App	IT
3	Recruitment System	HR
4	Payroll System	Finance
5	Marketing Campaign	Marketing
6	Database Migration	IT
7	Inventory System	[NULL]
8	CRM Implementation	Operations

2.

	AZ name	123 salary
1	Fajar Nugroho	0

3.

	AZ department

4.

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	AZ name ▼	AZ project ▼	123 hours_worked ▼	
1	Budi Santoso	Website Development	120.5	
2	Siti Rahayu	Website Development	95.5	

Praktikum 3: Fungsi Agregasi

Konsep Dasar

- **GROUP BY:** Mengelompokkan data berdasarkan kolom tertentu
- **HAVING:** Filter hasil agregasi (setelah GROUP BY)
- **Fungsi Agregasi:** COUNT, SUM, AVG, MAX, MIN

Latihan Praktis

1. Basic Agregasi

```
SELECT
    COUNT(*) as total_employees,
    AVG(salary) as avg_salary,
    MAX(salary) as max_salary
FROM employees;
```

2. GROUP BY

```
SELECT
    d.name as department,
    COUNT(e.id) as employee_count,
    AVG(e.salary) as avg_salary
FROM departments d
LEFT JOIN employees e ON d.id = e.department_id
GROUP BY d.name;
```

3. HAVING

```
SELECT
    d.name as department,
    COUNT(e.id) as employee_count
FROM departments d
LEFT JOIN employees e ON d.id = e.department_id
GROUP BY d.name
HAVING COUNT(e.id) > 2;
```

4. Multiple Agregasi

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```
SELECT
    p.name as project,
    SUM(ep.hours_worked) as total_hours,
    COUNT(ep.employee_id) as total_employees,
    AVG(ep.hours_worked) as avg_hours_per_employee
FROM projects p
LEFT JOIN employee_projects ep ON p.id = ep.project_id
GROUP BY p.name;
```

Soal Latihan Praktikum 3

1. Tampilkan total budget projects per department
2. Tampilkan department dengan rata-rata gaji di atas 8.5 juta
3. Tampilkan project dengan total jam kerja lebih dari 200 jam
4. Tampilkan employee dengan total jam kerja terbanyak

Query Mahasiswa:

```
SELECT
    d.name as department,
    SUM(p.budget) as total_budget
FROM departments d
LEFT JOIN projects p ON d.id = p.department_id
GROUP BY d.name;

SELECT
    d.name as department,
    AVG(e.salary) as avg_salary
FROM departments d
LEFT JOIN employees e ON d.id = e.department_id
GROUP BY d.name
HAVING AVG(e.salary) > 8500000;

SELECT
    p.name as project,
    SUM(ep.hours_worked) as total_hours
FROM projects p
LEFT JOIN employee_projects ep ON p.id = ep.project_id
GROUP BY p.name
HAVING SUM(ep.hours_worked) > 200;

SELECT
    e.name,
    SUM(ep.hours_worked) as total_hours
FROM employees e
LEFT JOIN employee_projects ep ON e.id = ep.employee_id
GROUP BY e.name
ORDER BY total_hours DESC
LIMIT 1;
```

Hasil Eksekusi:

1.

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	AZ department	123 total_budget
1	Marketing	60,000,000
2	Operations	55,000,000
3	Finance	45,000,000
4	IT	160,000,000
5	HR	30,000,000

2.

	AZ department	123 avg_salary
1	Finance	9,000,000
2	HR	8,750,000

3.

	AZ project	123 total_hours
1	Payroll System	620.5
2	Website Development	216
3	Recruitment System	380.5
4	Mobile App	290
5	Marketing Campaign	335.5

4.

	AZ name	123 total_hours
1	Siti Rahayu	245.5

Praktikum 4: Common Table Expression (CTE)

Konsep Dasar

- **CTE:** Tabel sementara yang hanya ada selama eksekusi query
- **Keunggulan:** Lebih mudah dibaca dan dimaintain dibanding subquery
- **Syntax:** WITH cte_name AS (SELECT ...) SELECT * FROM cte_name

Latihan Praktis

1. Basic CTE

```
WITH high_salary_employees AS (  
    SELECT * FROM employees WHERE salary > 8500000  
)  
SELECT * FROM high_salary_employees ORDER BY salary DESC;
```

2. Multiple CTEs

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```
WITH dept_stats AS (  
    SELECT  
        department_id,  
        COUNT(*) as emp_count,  
        AVG(salary) as avg_salary  
    FROM employees  
    GROUP BY department_id  
)  
project_stats AS (  
    SELECT  
        department_id,  
        COUNT(*) as project_count  
    FROM projects  
    GROUP BY department_id  
)  
SELECT  
    d.name as department,  
    ds.emp_count,  
    ds.avg_salary,  
    ps.project_count  
FROM departments d  
LEFT JOIN dept_stats ds ON d.id = ds.department_id  
LEFT JOIN project_stats ps ON d.id = ps.department_id;
```

3. CTE vs Subquery

```
-- Subquery (lebih rumit)  
SELECT name, salary  
FROM employees  
WHERE salary > (SELECT AVG(salary) FROM employees);  
  
-- CTE (lebih mudah dibaca)  
WITH avg_salary AS (  
    SELECT AVG(salary) as avg_sal FROM employees  
)  
SELECT name, salary  
FROM employees, avg_salary  
WHERE salary > avg_sal;
```

4. CTE Complex Query

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```
WITH employee_project_stats AS (  
    SELECT  
        e.name as employee_name,  
        COUNT(ep.project_id) as total_projects,  
        SUM(ep.hours_worked) as total_hours  
    FROM employees e  
    LEFT JOIN employee_projects ep ON e.id = ep.employee_id  
    GROUP BY e.name  
)  
SELECT  
    employee_name,  
    total_projects,  
    total_hours,  
    CASE  
        WHEN total_hours > 200 THEN 'High'  
        WHEN total_hours > 100 THEN 'Medium'  
        ELSE 'Low'  
    END as workload_level  
FROM employee_project_stats  
ORDER BY total_hours DESC;
```

Soal Latihan Praktikum 4

1. Buat CTE untuk menampilkan top 3 employee dengan gaji tertinggi
2. Buat CTE untuk menghitung rata-rata jam kerja per department
3. Buat CTE untuk menampilkan project dengan budget di atas rata-rata
4. Buat CTE berjenjang (multiple CTEs) untuk analisis department lengkap

Query Mahasiswa:

1.

```
WITH top_employees AS (  
    SELECT name, salary  
    FROM employees  
    ORDER BY salary DESC  
    LIMIT 3  
)  
SELECT * FROM top_employees;
```

2.

```
WITH dept_avg_hours AS (  
    SELECT  
        d.name as department,  
        AVG(ep.hours_worked) as avg_hours  
    FROM departments d  
    LEFT JOIN employees e ON d.id = e.department_id  
    LEFT JOIN employee_projects ep ON e.id = ep.employee_id  
    GROUP BY d.name  
)  
SELECT * FROM dept_avg_hours;
```

3.

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```
WITH avg_budget AS (  
    SELECT AVG(budget) as avg_bud FROM projects  
)  
SELECT p.name, p.budget  
FROM projects p, avg_budget  
WHERE p.budget > avg_budget.avg_bud;
```

4.

```
WITH emp_summary AS (  
    SELECT  
        department_id,  
        COUNT(*) as emp_count,  
        AVG(salary) as avg_salary  
    FROM employees  
    GROUP BY department_id  
)  
project_summary AS (  
    SELECT  
        department_id,  
        COUNT(*) as project_count,  
        SUM(budget) as total_budget  
    FROM projects  
    GROUP BY department_id  
)  
hours_summary AS (  
    SELECT  
        e.department_id,  
        SUM(ep.hours_worked) as total_hours  
    FROM employees e  
    LEFT JOIN employee_projects ep ON e.id = ep.employee_id  
    GROUP BY e.department_id  
)  
SELECT  
    d.name as department,  
    es.emp_count,  
    es.avg_salary,  
    ps.project_count,  
    ps.total_budget,  
    hs.total_hours  
FROM departments d  
LEFT JOIN emp_summary es ON d.id = es.department_id  
LEFT JOIN project_summary ps ON d.id = ps.department_id  
LEFT JOIN hours_summary hs ON d.id = hs.department_id;
```

Hasil Eksekusi:

1.

	A-Z name	123 salary
1	Rudi Hermawan	9,500,000
2	Joko Widodo	9,200,000
3	Ahmad Fauzi	9,000,000

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	A-Z department	123 avg_hours
1	Marketing	167.75
2	Operations	180
3	Finance	206.833333333333
4	IT	112.75
5	HR	190.25

3.

	A-Z name	123 budget
1	Website Development	50,000,000
2	Mobile App	75,000,000
3	Marketing Campaign	60,000,000
4	CRM Implementation	55,000,000

4.

	A-Z department	123 emp_count	123 avg_salary	123 project_count	123 total_budget	123 total_hours
1	IT	3	8,233,333.33333333	3	160,000,000	676.
2	HR	2	8,750,000	1	30,000,000	380.
3	Finance	3	9,000,000	1	45,000,000	620.
4	Marketing	2	8,000,000	1	60,000,000	335.
5	Operations	1	6,500,000	1	55,000,000	18

Tugas Akhir Integrasi

Buat query lengkap yang menggabungkan semua konsep untuk menjawab pertanyaan berikut:

1. Tampilkan 5 department dengan total gaji tertinggi
2. Tampilkan project dengan rata-rata jam kerja per employee tertinggi
3. Tampilkan employee yang bekerja di lebih dari 1 project
4. Buat laporan performance department: nama department, jumlah employee, jumlah project, total budget projects, rata-rata gaji, total jam kerja

Query Mahasiswa:

1.

```
SELECT
    d.name as department,
    SUM(e.salary) as total_salary
FROM departments d
LEFT JOIN employees e ON d.id = e.department_id
GROUP BY d.name
ORDER BY total_salary DESC
LIMIT 5;
```

2.

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```
SELECT
    p.name as project,
    AVG(ep.hours_worked) as avg_hours_per_employee
FROM projects p
JOIN employee_projects ep ON p.id = ep.project_id
GROUP BY p.name
ORDER BY avg_hours_per_employee DESC
LIMIT 1;
```

3.

```
SELECT |
    e.name,
    COUNT(ep.project_id) as jumlah_project
FROM employees e
JOIN employee_projects ep ON e.id = ep.employee_id
GROUP BY e.name
HAVING COUNT(ep.project_id) > 1;
```

4.

```
WITH emp_stats AS (
    SELECT
        department_id,
        COUNT(*) as jumlah_employee,
        AVG(salary) as rata_rata_gaji
    FROM employees
    GROUP BY department_id
),
project_stats AS (
    SELECT
        department_id,
        COUNT(*) as jumlah_project,
        SUM(budget) as total_budget
    FROM projects
    GROUP BY department_id
),
hours_stats AS (
    SELECT
        e.department_id,
        SUM(ep.hours_worked) as total_jam_kerja
    FROM employees e
    LEFT JOIN employee_projects ep ON e.id = ep.employee_id
    GROUP BY e.department_id
)
SELECT
    d.name as nama_department,
    COALESCE(es.jumlah_employee, 0) as jumlah_employee,
    COALESCE(ps.jumlah_project, 0) as jumlah_project,
    COALESCE(ps.total_budget, 0) as total_budget_projects,
    COALESCE(es.rata_rata_gaji, 0) as rata_rata_gaji,
    COALESCE(hs.total_jam_kerja, 0) as total_jam_kerja
FROM departments d
LEFT JOIN emp_stats es ON d.id = es.department_id
LEFT JOIN project_stats ps ON d.id = ps.department_id
LEFT JOIN hours_stats hs ON d.id = hs.department_id
ORDER BY d.name;
```

Hasil Eksekusi:

1.

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	AZ department	123 total_salary
1	Finance	27,000,000
2	IT	24,700,000
3	HR	17,500,000
4	Marketing	16,000,000
5	Operations	6,500,000

2.

	AZ project	123 avg_hours_per_employee
1	Payroll System	206.833333333333

3.

	AZ name	123 jumlah_project
1	Joko Widodo	2
2	Siti Rahayu	2
3	Budi Santoso	2

4.

	AZ nama_department	123 jumlah_employee	123 jumlah_project	123 total_budget_projects	123 rata_rata_gaji	123 total_jam_kerja
1	Finance	3	1	45,000,000	9,000,000	620.5
2	HR	2	1	30,000,000	8,750,000	380.5
3	IT	3	3	160,000,000	8,233,333.33333333	676.5
4	Marketing	2	1	60,000,000	8,000,000	335.5
5	Operations	1	1	55,000,000	6,500,000	180.5

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Refleksi Pembelajaran

Nama Mahasiswa: _____

NIM: _____

Hal yang sudah dipahami:

- SELECT dengan LIMIT/OFFSET
- Berbagai jenis JOIN
- Fungsi agregasi dengan GROUP BY dan HAVING
- Penggunaan CTE untuk query kompleks

Kesulitan yang dialami: